Coding Practice Set - 05

Basic String Metrics & Character Counting

1. **Write a program to calculate the length of a string without using any built-in library functions.**
2. **Write a function to calculate the length of a string.**
3. **Write a program to count the occurrence of a given character in a given string.**
4. **Write a program to count vowels in a given string.**
5. **Write a program to count the total number of consonants, digits, and special characters in a given string.**
6. **Write a program to find the highest frequency character in a given string.**

Case Conversions & Transformations

7. **Write a program to convert a given string into uppercase.**
8. **Write a program to convert a given string into lowercase.**
9. **Write a program to toggle the case of each character in a string (convert uppercase to lowercase and vice versa).**
10. **Write a program to capitalize the first letter of each word in a given sentence.**

String Reversals & Comparisons

11. **Write a program to reverse a string.**
12. **Write a function to reverse a string.**

13. **Write a function to compare two strings.**
14. **Write a program to check whether a given string is a palindrome (reads the same forward and backward).**
15. **Write a program to reverse the order of words in a given sentence (e.g., "Hello World" becomes "World Hello").**

Advanced Manipulation & Searching

16. **Write a program to remove all spaces from a given string.**
17. **Write a program to find the first occurrence of a substring within a main string (similar to the strstr logic).**
18. **Write a program to check if two strings are anagrams of each other (contain the exact same characters in a different order).**
19. **Write a program to extract a substring from a given string given a starting index and a length.**

Multi-String Handling & 2-D Arrays

20. **Write a program to sort 10 city names stored in a two-dimensional array, taken from the user.**
21. **Write a program to search for a specific city name within a list of 5 user-entered city names stored in a 2-D character array.**
22. **Write a program to count the total number of vowels across a collection of 5 different strings stored in a 2-D array**